Rejoin VM to Domain using CSE_RDP SSH

Last updated by | Clayton Fuselier | Jun 29, 2025 at 7:12 AM GMT+5:30

Tags

cw.How-To	cw.RDP-SSH	cw.Reviewed-09-2022

Contents

Summary

In some cases you will not have the option to access a VM remotely and the trusted communication between your machine and the domain is broken and you need to rebuild this trust. One way to do it is rejoining the machine to the domain. This is useful in scenarios where you don't have access to a VM either with local or domain IDs and you want to add this access back using GPO but the trusted AD channel of this machine to the domain is broken.

Customer Enablement

https://docs.microsoft.com/en-us/troubleshoot/azure/virtual-machines/troubleshoot-broken-secure-channel#remove-the-vm-from-the-domain-and-re-join-the-domain

Limitations

1. This extension will work only if the VM has connectivity as this feature uses the same pipeline as Custom Script Extension
2. As any other extension, this extension will work if only the Guest Agent is installed and is working as expected
3. Only the very first time the extension is used, will inject the script to execute. If you use this feature later on, the extension will identified it was already used and will not upload the new script to execute. To avoid this, you will need to ensure the extension is not installed on the VM prior to its use
4. The minimum time to run a script is about 20 seconds
5. The maximum time a script can run is 90 minutes, after which it will time out

Instructions

Using a Domain Administrator Account

Using **CustomScriptExtension**

Basically you're going to build 2 scripts, to leverage the Custom Script Extension feature, via portal:

1.1. Create a script call *Unjoin.ps1* file with this content, and deploy it as a Custom Script Extension on the Azure Portal:

```
cmd /c "netdom remove <<MachineName>> /domain:<<DomainName>> /userD:<<DomainAdminhere>> /passwordD:<<PasswordHere>> /reboot:10 /Force"
```

1.2. This script will take the VM out of the domain forcefully and will restart after 10sec. Ask the customer to involve his AD team to clean up the Computer object on the Domain side.

1.3. Once the cleanup is done, proceed to rejoin this VM. Create the script called *JoinDomain.ps1* with this content and deploy it as a Custom Script Extension on the Azure Portal:

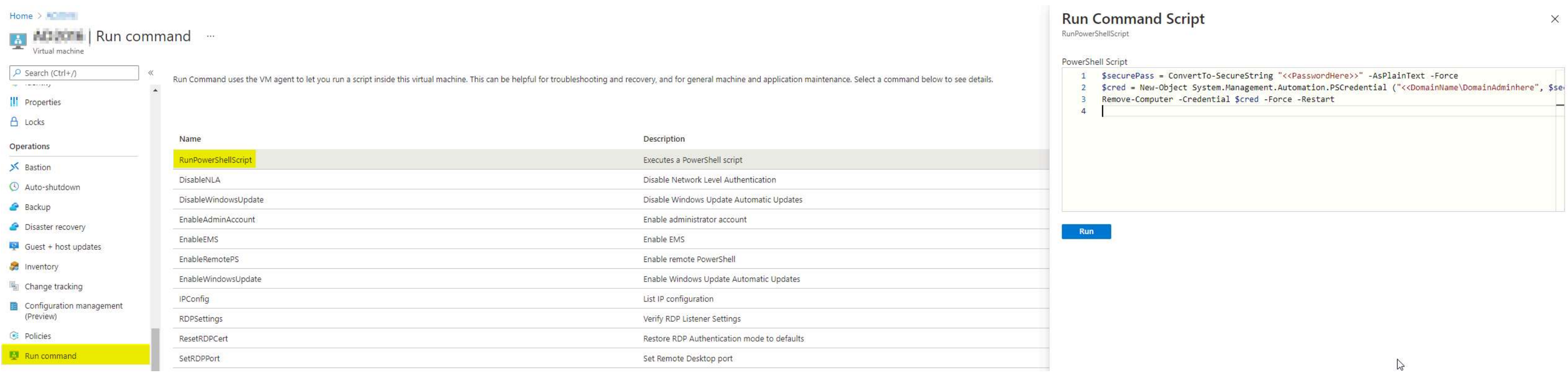
```
cmd /c "netdom join <<MachineName>> /domain:<<DomainName>> /userD:<<DomainAdminhere>> /passwordD:<<PasswordHere>> /reboot:10"
```

1.4. This will join the VM on the domain using the specified credentials – Please double check the domain name and user credentials

Disclaimer: Note that this method works only for Windows Server. *Netdom* is not installed by default on Windows Client.

Using **Run Command**

2.1. From Azure Portal, navigate to the VMBlade, select *Run command* and *RunPowershellScript*. On the *Run Command Script* edit then run the following script:




```
# Remove VM from domain
$securePass = ConvertTo-SecureString "<<PasswordHere>>" -AsPlainText -Force
$cred = New-Object System.Management.Automation.PSCredential ("<<DomainName\DomainAdminhere>>", $securePass)
Remove-Computer -Credential $cred -Force -Restart
```

2.2. This script will take the VM out of the domain and will restart it.

2.3. Once the VM has been rebooted, wait for the OS to boot and confirm that the *Guest Agent* has started. Now, by following the same procedure mentioned at step 2.1, run the following script to join the VM back to the domain:

```
# Join VM to domain
$securePass = ConvertTo-SecureString "<<PasswordHere>>" -AsPlainText -Force
$cred = New-Object System.Management.Automation.PSCredential ("<<DomainName\DomainAdminhere>>", $securePass)
Add-Computer -DomainName "<<DomainName>>" -Credential $cred -Force -Restart
```

If the above steps fail for whatever reason, this method can also be attempted with the advantage of needing a single reboot after the disjoin and rejoin instead of 2:

```
# Enter variables (instruct the cx not to screen share while entering the password and to not screenshare until the pw has been cleared)
$adminAccount = "azureadmin"
$adminPassword = "Welcome2Azure!"
$domain = "corp.contoso.com"
clear

# Run these commands
$adComputer = Get-WmiObject Win32_ComputerSystem
$adComputer.UnjoinDomainOrWorkGroup($adminPassword, $adminAccount, 0)
$adComputer.JoinDomainOrWorkGroup($domain, $adminPassword, $adminAccount, $null, 3)
Restart-Computer
```

Using **Serial Console**

▶ Details

Using a Local Administrator Account

If the customer doesn`t have access to a Domain Administrator account, proceed with the following steps to remove the VM from domain.

Disclaimer: This method can be used only via *Run Command* as the VM needs to be completely isolated from the rest of the environment.

1. Confirm via ASC or Azure Portal that the VM has a NSG attached to the *NIC*:

Inbound port rules | Outbound port rules | Application security groups | Load balancing

Network security group [illegible] (attached to subnet: [illegible])
Impacts 1 subnets, 0 network interfaces

Priority	Name	Port	Protocol
2700	AllowCorpnet	Any	Any
2701	AllowSAW	Any	Any
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalancerInBound	Any	Any
65500	DenyAllInBound	Any	Any

Network security group [illegible] (attached to network interface: [illegible])
Impacts 0 subnets, 1 network interfaces

Priority	Name	Port	Protocol
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalancerInBound	Any	Any
65500	DenyAllInBound	Any	Any

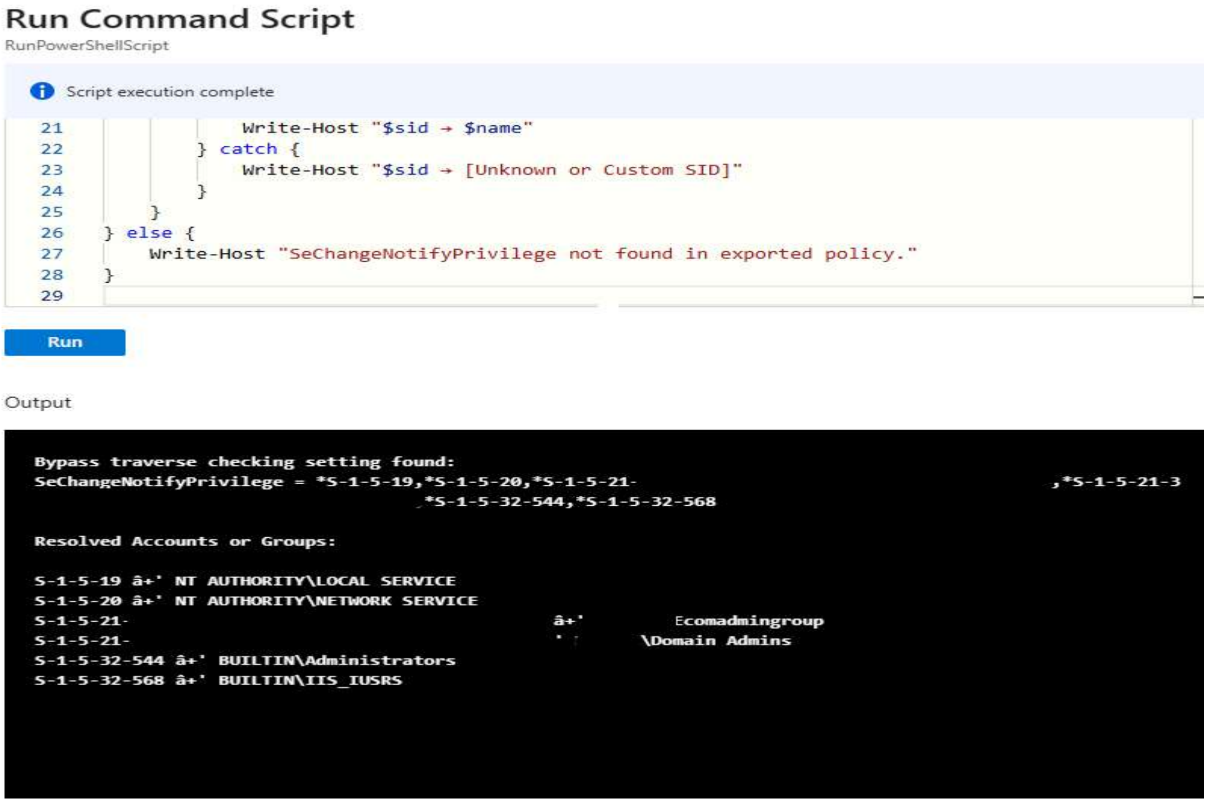
1.1. If there is no NSG attached to the NIC, proceed with creating one in the same *ResourceGroup* and *Region* and attach the NIC of the affected VM to it.
If you plan to use an existing NSG, make sure that it is not attached to any other VM/NIC before adding the rules.

1.2. Create Inbound/Outbound traffic rules with the highest priority, to deny All Access

2. Once the VM has been isolated, navigate to *Run Command* and run the following script after entering valid credentials:

```
$userName = "machinename\localadmin"
$userPassword = "localadminpassword"
$secStringPassword = ConvertTo-SecureString $userPassword -AsPlainText -Force
$credObject = New-Object System.Management.Automation.PSCredential ($userName, $secStringPassword)
Remove-Computer -UnjoinDomaincredential $credObject -PassThru -Verbose -Restart -Force
```

Wait for the output that confirms the command completed successfully before leaving/refreshing the *Run Command* page



Once the above operation has been completed, remove the NSG rules created previously or detach the NSG if you created one, connect to the VM by using the local administrator account and join the VM back manually if needed.

Need additional help or have feedback?

To engage the Azure RDP-SSH SMEs...	*To provide feedback on this page...*	*To provide kudos on this page...*
If you are in the **OneVM team** then reach out to the **RDP-SSH SMEs** AVA channel via Teams. If you are a **VM Delivery Partner** then follow your team's process for swarming or ask your TA. Make sure to use the **Ava process** for faster assistance.	Use the **RDP-SSH Feedback** form to submit detailed feedback on improvements or new content ideas for RDP-SSH. ***Please note*** the link to the page is required when submitting feedback on existing pages! If it is a new content idea, please put N/A in the Wiki Page Link.	Use the **RDP-SSH Kudos** form to submit kudos on the page. Kudos will help us improve our wiki content overall! ***Please note*** the link to the page is required when submitting kudos!